

TEAM STRUCTURE

Our team consists of 13 students and 2 faculty advisors working across two simultaneous missions. For the Rocket Challenge, Maximiliano Funoy serves as Mission Leader, José de J. Campos as Safety Manager, Luis H. Mota as Operations Manager, and Jean Macías as Technology Manager. The Satellite Challenge is led by Emilio Guadarrama, with Iliana Cosette Saldaña as Safety Manager, Braulio García as Operations Manager, and Karla G. Morales as Technology Manager. Both missions are guided by faculty advisors José L. Henríquez and Marco A. Alcaráz, along with team members Damaris Ramos, Sofía Pacheco, Nicolás Infante, Donghyun Kim, and Diego A. Gutiérrez.

SYSTEMS ENGINEERING

We're tackling both missions at once, which has been great for sharing hardware, a unified communication system, and a single testing schedule. Our flight simulations are on track for the 1,000 m AGL apogee target. We ran finite element analysis on our key structures and confirmed solid safety margins, while CFD simulations validated our aerodynamic design. We've also physically tested our structural materials to confirm they hold up under expected flight conditions. IGNIS-1's primary mission focuses on detecting forest fires by gathering temperature, humidity, pressure, gas concentration, and particulate matter data. It also has a secondary mission to assess radio link quality at various altitudes, no extra hardware needed since everything runs on the shared flight computer.

BUDGET AND FUNDRAISING

Our Science Department at Tecnológico de Monterrey has covered all project development costs so far. The CanSat budget is comfortably within competition limits. For travel, we're footing the bill ourselves, about 11,000 MXN per person flying with LATAM Airlines. We're talking with the Jalisco state government and a few private companies to secure sponsorship for travel and registration fees, and we're gearing up for some fundraising events on campus too.

TRAVEL PLANS

All 15 of us fly out of Guadalajara on August 26. We spend the first days in Rio de Janeiro handling arrival logistics, checking equipment, meeting with local universities, and collaborating on propellant work in their lab. On August 30 we head to São Paulo to restock and regroup, then drive to Iacanga on September 1 for check-in. Competition runs September 2–5, with a buffer day in São Paulo on September 6 before heading home on September 7. We're booking AirBnBs in each city and conversations with hosts are going smoothly. For getting around, we're renting one or two vans for the whole team and all our gear.

TECHNICAL PROJECT UPDATES

Rocket — Design and Manufacturing

The airframe is complete, with a few minor details left to finish. The motor casing, combustion chamber, and nozzle are manufactured, and we're wrapping up the junctions and nose cone. We have everything ready for recovery (drogue, main parachute and deployment mechanism) and assembly is underway. Our backup commercial altimeter has been tested and is good to go. The first version of our SRAD flight computer is fully assembled and operational. We're now in the layout phase for the next version, adding more sensors, dual pyrotechnic control, and long-range telemetry, targeting integration before our June flight test.

Rocket — Testing and Launch Operations

Our first motor static fire is scheduled for May 30th, where we'll measure actual thrust against our simulations. The full vehicle flight test follows on June 22nd, covering stability, avionics, and recovery deployment all at once. On the propellant front, we've reached out to several Brazilian universities about producing the grains locally before the event and have Zoom meetings lined up, no confirmation yet, but it's our top priority and we're working hard to close it.

CanSat IGNIS-1

All sensors for both missions are in hand and tested on a prototype circuit, everything is working as expected. The radio link checks out too. The shared flight computer PCB design is still in progress, and the physical CanSat structure hasn't been built yet; we're aiming for around 300 g, but a prototype is being done using an esp32 and combining different environmental and communication sensors. Our ground station software is being developed in-house and will show live sensor readings, graphs of altitude and other key parameters, and a map tracking both the rocket and the CanSat simultaneously.

Safety

We've examined each subsystem (propulsion, avionics, and recovery) and identified potential issues along with response strategies. All circuits have been bench tested, and we ran burn tests on our propellant to confirm it performs as expected. Everyone handling pyrotechnic devices has reviewed the necessary safety procedures ahead of our upcoming tests.

PITCH VIDEO:

Tomas rápidas de sensores, soldadura, cohete, cad, pcb (altium), equipo trabajando.

Every year, wildfires destroy thousands of hectares of forest, during those disasters plenty of people lose communication separating families from their loved ones.

Grabacion de alguien hablando diciendo esto:

What if aerospace technology could help solve such problems ?

Otra persona:

We are not just building a rocket, we are building a complete and functional mission.

Reuniones, miembros del equipo trabajando

Who are we?

Fotos de los integrantes

We are a team of 24 students and 5 faculty advisors from Tecnológico de Monterrey. Engineers, programmers, designers and problem solvers working together to accomplish a goal in common. To represent Mexico in LASC 2026.

Más detalles de simulaciones, diseño, aviónica, estación terreno, etc.

The Challenge:

This year we are simultaneously developing both a high-powered rocket and a Cansat mission.

By integrating systems, testing schedules and communications architecture, we have optimized our development process as a single aerospace mission. Our rocket is designed to reach an apogee of 1,000 meters above ground level.

While CFD simulations validated our aerodynamic design. We've also physically tested our structural materials to confirm they hold up under expected flight conditions.

Incendios, graficas, maps, humo, comunicación, bosques y zonas secas.

Mission:

IGNIS-1's primary mission focuses on detecting the conditions that lead forest fires by gathering temperature, humidity, pressure, gas concentration, and particulate matter data.

It also has a secondary mission to assess radio link quality at various altitudes and with this analyze how rural zones lose communication in plenty of areas.

No extra hardware needed since everything runs on the shared flight computer.

Testing and security:

Test de propelentes y lanzamientos

.Our structural components were analyzed through finite element simulations and physically tested to ensure safety during flight."

We developed our own SRAD flight computer, telemetry systems, and avionics architecture. Right now, we are preparing for our upcoming static fire and full flight tests.

Safety:

Safety is one of our top priorities. As we move toward launch, every test brings us one step closer to competition."

Mapa de gdl a brasil bandera mexicana

Cierre:

In September 2026, we are going to Brazil.. For us, this project represents much more than a competition. It represents innovation, collaboration, and the opportunity to create aerospace technology with real impact.

We are IGNITIA, a student team driven by curiosity, engineering, and ambition.